

Appendix A. Generator-Level Unit Commitment Formulation

This appendix presents the exact mathematical structure of the final generator-level unit commitment (UC) model used in the project. The model is solved on a rolling horizon and co-optimizes generator dispatch, thermal commitment decisions, renewable allocation, and battery storage operation.

A.1 Sets and Indices

- $t \in \mathcal{T}$: hourly time periods in the rolling horizon
- $g \in \mathcal{G}$: all dispatchable generators
- $\mathcal{G}^{bin} \subseteq \mathcal{G}$: binary commitment generators (GAS, BIOFUEL)
- $\mathcal{G}^{cont} \subseteq \mathcal{G}$: continuous dispatch generators (HYDRO, NUCLEAR)

A.2 Parameters

For each hour t :

- D_t : Ontario demand (MW)
- R_t : renewable energy input available to the UC (MW)
- S_t : stress-hour cost adder
- $\bar{P}^{stor}$: battery power limit (MW)
- $\bar{E}^{stor}$: battery energy limit (MWh)

For each generator g :

- $\bar{P}_{g,t}$: maximum output (MW)
- $\underline{P}_{g,t}$: minimum output (MW)
- c_g^{var} : variable production cost (\$/MWh)
- c_g^{su} : startup cost (\$)
- c_g^{sd} : shutdown cost (\$)

- RU_g : ramp-up limit (MW/h)
- RD_g : ramp-down limit (MW/h)
- MU_g : minimum up-time (hours)
- MD_g : minimum down-time (hours)

Battery parameters:

- η^{ch} : charging efficiency
- η^{dis} : discharging efficiency
- E^{init} : initial battery state of charge
- E^{final} : required final state of charge

Penalty parameters:

- c^{curt} : renewable curtailment penalty
- c^{shed} : load-shedding penalty
- c^{over} : overgeneration penalty
- c^{thr} : battery-throughput penalty

A.3 Decision Variables

Renewable and system balancing:

- $r_t^{dir} \geq 0$: renewable power used directly to serve load
- $r_t^{curt} \geq 0$: curtailed renewable power
- $l_t^{shed} \geq 0$: load shed
- $o_t^{over} \geq 0$: overgeneration slack

Battery:

- $ch_t \geq 0$: battery charging power
- $dis_t \geq 0$: battery discharging power
- $e_t \geq 0$: battery state of charge
- $m_t \in \{0, 1\}$: battery charge/discharge mode indicator

Generators:

- $p_{g,t} \geq 0$: generator dispatch
- $u_{g,t} \in \{0,1\}$: commitment state for $g \in \mathcal{G}^{bin}$
- $su_{g,t} \in \{0,1\}$: startup indicator for $g \in \mathcal{G}^{bin}$
- $sd_{g,t} \in \{0,1\}$: shutdown indicator for $g \in \mathcal{G}^{bin}$

A.4 Objective Function

The model minimizes total operating cost over the rolling horizon:

$$\min \sum_{t \in \mathcal{T}} \left[\sum_{g \in \mathcal{G}} (c_g^{var} + S_t) p_{g,t} + \sum_{g \in \mathcal{G}^{bin}} (c_g^{su} su_{g,t} + c_g^{sd} sd_{g,t}) \right. \\ \left. + c^{curt} r_t^{curt} + c^{shed} l_t^{shed} + c^{over} o_t^{over} + c^{thr} ch_t \right] \quad (1)$$

Interpretation:

- generator dispatch incurs variable production cost
- binary thermal units also incur startup and shutdown costs
- curtailment, load shedding, and overgeneration are penalized
- a small throughput penalty discourages unnecessary battery cycling
- the stress-hour adder increases the model's incentive to preserve flexibility in difficult hours

A.5 Constraints

1. Renewable allocation

$$r_t^{dir} + r_t^{curt} + ch_t = R_t \quad \forall t \quad (2)$$

2. System power balance

$$r_t^{dir} + dis_t + \sum_{g \in \mathcal{G}} p_{g,t} + l_t^{shed} - o_t^{over} = D_t \quad \forall t \quad (3)$$

3. Battery charge/discharge exclusivity

$$ch_t \leq \bar{P}^{stor} m_t \quad \forall t \quad (4)$$

$$dis_t \leq \bar{P}^{stor} (1 - m_t) \quad \forall t \quad (5)$$

4. Battery state of charge dynamics

$$e_t = e_{t-1} + \eta^{ch} ch_t - \frac{1}{\eta^{dis}} dis_t \quad \forall t \quad (6)$$

For the first modeled hour:

$$e_1 = E^{init} + \eta^{ch} ch_1 - \frac{1}{\eta^{dis}} dis_1 \quad (7)$$

Battery bounds:

$$0 \leq e_t \leq \bar{E}^{stor} \quad \forall t \quad (8)$$

Terminal condition:

$$e_{|\mathcal{T}|} = E^{final} \quad (9)$$

5. Generator output limits for binary units

For $g \in \mathcal{G}^{bin}$:

$$p_{g,t} \leq \bar{P}_{g,t} u_{g,t} \quad \forall t \quad (10)$$

$$p_{g,t} \geq \underline{P}_{g,t} u_{g,t} \quad \forall t \quad (11)$$

6. Commitment state transition

For $g \in \mathcal{G}^{bin}$:

$$u_{g,t} - u_{g,t-1} = su_{g,t} - sd_{g,t} \quad \forall t \quad (12)$$

Also:

$$su_{g,t} + sd_{g,t} \leq 1 \quad \forall t \quad (13)$$

7. Ramp constraints

For binary units:

$$p_{g,t} - p_{g,t-1} \leq RU_g + \bar{P}_{g,t} su_{g,t} \quad \forall g \in \mathcal{G}^{bin}, t \quad (14)$$

$$p_{g,t-1} - p_{g,t} \leq RD_g + \bar{P}_{g,t-1} sd_{g,t} \quad \forall g \in \mathcal{G}^{bin}, t \quad (15)$$

For continuous units:

$$p_{g,t} - p_{g,t-1} \leq RU_g \quad \forall g \in \mathcal{G}^{cont}, t \quad (16)$$

$$p_{g,t-1} - p_{g,t} \leq RD_g \quad \forall g \in \mathcal{G}^{cont}, t \quad (17)$$

Special continuous-unit ramp assumptions:

- NUCLEAR: 1% of rated power per minute, or 60% of rated power per hour
- HYDRO: allowed to move from 0 to P_{max} within the hour

8. Minimum up-time constraints

$$\sum_{\tau=t-MU_g+1}^t su_{g,\tau} \leq u_{g,t} \quad \forall g \in \mathcal{G}^{bin}, t \quad (18)$$

9. Minimum down-time constraints

$$\sum_{\tau=t-MD_g+1}^t sd_{g,\tau} \leq 1 - u_{g,t} \quad \forall g \in \mathcal{G}^{bin}, t \quad (19)$$

A.6 Policy Interpretation in the Project

The same UC structure is used for multiple policy comparisons, with only the renewable-input definition changing:

- `historical_actual`: observed storage and observed system operation baseline
- `no_storage_uc`: UC with battery disabled
- `ieso_forecast_uc`: UC using raw IESO renewable forecast
- `forecast_informed_uc`: UC using the project’s best ML renewable forecast
- `perfect_foresight_uc`: UC using actual renewable output

A.7 Modeling Notes

- the model is system-level, not nodal
- no transmission constraints are included
- no reserve constraints are included
- no voltage or network feasibility constraints are included
- storage is charged from modeled renewable energy
- the rolling horizon is solved in sequential blocks, with commitment state carried forward between blocks
- the final main case uses a 168-hour rolling horizon

A.8 Practical Meaning

This formulation is more physically realistic than a simple storage-only dispatch model because it embeds battery decisions inside a full UC environment with thermal commitment logic, startup and shutdown costs, minimum stable generation, ramping limits, minimum up/down requirements, explicit state-of-charge tracking, and hourly system balance.